Junhao Zhao

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2027 Research Internships | Credit Assignment & Long-Horizon Learning

Available May 21 - September 1, 2027; flexible start and end dates.

Research interests: I study general questions of autoregressive modeling and long-horizon learning in time series and brain dynamics, with a focus on credit assignment and how models use history to predict evolving states. More broadly, I am interested in how these principles can support learning in world models, reinforcement learning, and long-horizon agents.

Additional research: My previous projects include diffusion models for test-time adaptation and contrastive learning for human activity recognition.

Publications: CVPR 2025; PerCom 2026 (first author). **Preprint:** arXiv 2026 (first author).

Education

University of Maryland, College Park Sep 2025 - May 2030 (expected)

Ph.D. in Electrical and Computer Engineering | Advisor: Nan Xu

Tsinghua University Sep 2021 - Jul 2025

B.S. in Mathematics and Physics; B.S. in Mechanical Engineering | GPA: 3.67/4.00

University of Georgia Aug - Dec 2023

Exchange student | GPA: 4.00/4.00

Selected Research: Long-Horizon Learning & State Transitions

Research lead on both projects, from problem formulation through implementation and evaluation.

Large Distant Gradients Need Not Be Reliable: reliability-weighted credit assignment for long-horizon autoregressive forecasting

Junhao Zhao, David Michael Simberg, Jacob Kang, Colin Connor Kurniawan, Nan Xu.

arXiv preprint, 2026. arXiv:2609.12890 | Code Jun - Aug 2026

- **Learning signal:** Formulated long-horizon credit assignment as balancing predictable gradient signal against noise; identified when large distant gradients are unreliable.
- **Method:** Derived bounded Wiener gains that minimize local risk under a signal-noise model; implemented Internal-DW to reweight backward routes while preserving forward rollouts and horizon losses.
- **Evaluation:** Built a known-SNR control and held-out gradient diagnostics; evaluated eight application testbeds to isolate the effects of history availability and noise-model mismatch.
- **Results:** Reduced relative L_2 error by **5.2%-13.8% versus full BPTT** on four history-dominated, weak-drive testbeds (three matched seeds); outperformed gradient clipping and Jacobian regularization on three, with similar performance on shear flow; outperformed validation-selected truncated BPTT on three.

Cross-Subject Brain-State Transition Prediction Model

Ongoing research project (unpublished). Jul 2026 - Present

- **Problem formulation:** Formulated testable hypotheses about how brain-state history supports autoregressive prediction and generalization to unseen individuals.
- **Experimental design and implementation:** Built a unified evaluation pipeline for competing models; designed matched comparisons and ablations to isolate sources of predictive gains.
- **Theory and interpretation:** Combined mathematical analysis, controlled simulations, and empirical diagnostics to explain model behavior and identify the limits of its predictive performance.

Selected Publications & Prior Research

Everything to the Synthetic: Diffusion-driven Test-time Adaptation via Synthetic-Domain Alignment

Jiayi Guo, **Junhao Zhao**, Chaoqun Du, Yulin Wang, Chunjiang Ge, Zanlin Ni, Shiji Song, Humphrey Shi, Gao Huang.

CVPR 2025. Paper | Code

Tsinghua University | Advisors: Gao Huang and Jiayi Guo Jan - Nov 2024

- **Diffusion-based adaptation:** Led classification and segmentation experiments for diffusion-driven synthetic-domain alignment, implementing and running the evaluation.
- **Robustness under distribution shift:** Reproduced established baselines and conducted comparative experiments across ImageNet-C, ImageNet-W, CIFAR-10-C, and PASCAL-VOC-C.
- **Diagnostics:** Conducted component ablations, hyperparameter studies, and data-stream sensitivity analyses to identify sources of performance gains.
- **Method refinement:** Proposed self-ensembling to stabilize predictions.

CARE: Contrastive Alignment for ADL Recognition from Event-Triggered Sensor Streams

Junhao Zhao, Zishuai Liu, Ruili Fang, Jin Lu, Linghan Zhang, Fei Dou.

PerCom 2026. Paper

University of Georgia | Advisor: Fei Dou Oct 2023 - Sep 2024

- **Sensor representation learning:** Designed and implemented the evaluation for activity recognition using complementary sequence and image representations of event-triggered sensor streams.
- **Method development:** Proposed sensor-layout features, low-frequency event filtering, and a combined supervised contrastive and cross-entropy objective.
- **Component validation:** Evaluated on Milan, Cairo, and Kyoto7; ablated preprocessing, image encoding, the loss function, and the two-branch architecture to isolate their contributions.

Technical Skills

Programming: Python, C++, MATLAB.

Frameworks and tools: PyTorch, Distributed Data Parallel (DDP), Slurm.

Research methods: Autoregressive forecasting, gradient diagnostics, diffusion-based adaptation, contrastive learning, controlled ablations, and robustness evaluation.